

(603) 617-9302
Durham, NH

Bocheng Cui

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SKILLS

Programming Languages Golang, C, Python, C++, Java
Tools Docker, Git, Bash, Mini-net, AWS, GCP, Azure, LaTeX, Markdown, HTML, Conda, R, SQL, Node.js, PHP, OpenWhisk
Others Distributed Systems, Algorithm, Data Structure, Networks, etc

RESEARCH INTEREST

Fault tolerance and edge systems, as well as provenance and replication in distributed/cloud environments.

PUBLICATIONS

- **Bocheng Cui**, Aleksey Charapko. "Simpler is Better: Revisiting Mencius State Machine Replication", *The 43rd International Symposium on Reliable Distributed Systems (SRDS)*, 2024. [PDF]
- Owen Hilyard, **Bocheng Cui**, Marielle Webster, Abishek Bangalore Muralikrishna, Aleksey Charapko. "Cloudy Forecast: How Predictable is Communication Latency in the Cloud?", *2024 33rd International Conference on Computer Communications and Networks (ICCCN)*, 2024. [PDF]

PROJECTS

- Provenance file system in distributed platform** 2024
- Designing and developing a provenance file system for cloud platforms, aimed at providing enhanced tracking and management functionalities for cloud environments.
- Mencius Revisited** 2023
- Conducted an in-depth study of state machine replication protocols, including Mencius, E-Paxos, and Paxos, with a focus on enhancing the Mencius protocol.
 - Implemented optimizations in the failure handling mechanisms of the Mencius protocol, significantly increased max throughput by improving the protocol.
- Cloud latency test** 2023
- Designed and implemented a comprehensive tool for testing latency in cloud environments.
 - Conducted thorough evaluations and studies on latency performance and stability across various Cloud Providers, including AWS, GCP, and Azure.
- Fault Tolerance Sharded K/V Storage System with MapReduce** 2021
- Engineered a fault-tolerant K/V storage system, implementing Raft for reliability, and a sharding algorithm with customizable shard placement for efficient large-scale data handling.
 - Developed a non-disruptive data migration algorithm and integrated Google MapReduce for enhanced data processing and analysis.

EXPERIENCE

- Teaching Assistant** Aug 2021 — Present
University of New Hampshire
Durham, NH
- Served as a teaching assistant for both graduate and undergraduate courses.
- Research Associate** Jun 2024 — Aug 2024
Center for Coastal and Ocean Mapping/NOAA-UNH Joint Hydrographic Center
Durham, NH
- Develop a converter from Tiny 1.2 to IHO SVG Schema / Translate python scripts from 2.7 to 3.6.
- Researching Assistant** May 2022 — Aug 2022
University of New Hampshire
Durham, NH
- Conducted evaluations and optimizations of existing consistency protocols, focusing on improving efficiency and reliability in distributed systems.
- Cloud Software Developer** May 2022 — Aug 2022
Center for Coastal and Ocean Mapping/NOAA-UNH Joint Hydrographic Center
Durham, NH
- Implementation and testing of cloud based software and architectures for hydrographic and exploration bathymetric mapping.

EDUCATION

Ph.D. in Computer Science, *University of New Hampshire* Since 2022
Master. in Computer Science, *University of New Hampshire* 2018 - 2022
Bachelor of Material Science, *Qingdao University of Science & Technology* 2014 - 2018

ACTIVITIES AND HONORS

UNH College of Engineering and Physical Sciences (CEPS) Dean Scholarship 2018 - 2022
Vice President of the Students Union at Qingdao University of Science and Technology 2015 - 2017